

CHE 494

Selected Topics in Chemical Engineering

1. Course number and name

CHE 494 Materials for Energy

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: Two, 75 min/wk. lectures; Engineering.

3. Instructor's or course coordinator's name

Sanjay Behura

4. Text book, title, author, publisher, year

Introduction to Solid State Physics, 8th Edition, Charles Kittel, Wiley, 2004.

a. Other supplemental materials

2D Materials: Properties and Devices, P. Avouris, T. F. Heinz, and T. Low;

Physics of Solar Cells, Peter Würfel;

Semiconductor Material and Device Characterization, Dieter K. Schroder, 3rd Edition, Wiley.

5. Specific course information

a. Brief description of the content of the course (catalog description)

The goal of this course is to prepare students with foundational understanding of the advanced materials including low-dimensional nanostructures, tools required to synthesize and characterize such materials, and their potential applications in photovoltaic energy conversion.

b. Prerequisites or co-requisites

Consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Understand the science of advanced materials
- Learn methods and techniques for advanced materials processing
- Learn 2D material characterization methods
- Understand principles of solar energy harvesting
- Synthesize and characterize advanced materials in the lab
- Write final project term paper and give a presentation

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.		X		
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	X			
(3)	an ability to communicate effectively with a range of audiences.			X	
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.				X
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

Science of Advanced Materials: Quantum Theory of Solids; Semiconductor Physics; Low-Dimensional Materials Science; Modifications of Materials Properties. (This part will focus on physics and chemistry of materials, their unique structure and intriguing properties).

Advanced Materials Processing: Synthesis Techniques for Advanced Material Thin Films: Physical Vapor Deposition and Chemical Vapor Deposition. (This part will focus on various strategies for the nucleation of materials and understanding the ensuing growth mechanisms).

Material Characterization: Analysis of Materials by Raman Spectroscopy, X-Ray Photoelectron Spectroscopy, Scanning Electron Microscopy, Transmission Electron Microscopy, Scanning Probe Microscopy, and Electrical Characterization. (This part will focus on basic working principles of advanced characterization tools).

Principles of Solar Energy Harvesting: Properties of Sunlight; Solar Cell Operation; Design and Development of Advanced Solar Cells: Silicon Homojunction and Graphene-Based Schottky-Junction Solar Cells, Organic and Dye-Sensitized Molecular Solar Cells, and Hybrid Organic-Inorganic Perovskite Photovoltaic Cells.

Energy Lab Work: (i) Synthesis and Characterization of Low-Dimensional Materials (2D-Graphene and 1D-ZnO) and (ii) Design and Development of Dye-Sensitized Molecular Cells.